PAPER_861: Kepler Orrery V 35-Frame Iterative U_b Model Refinement

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** KeplerOrreryV35FrameIterativeUbCalc (CP4 #445) **CVW:** v2.0.0 compliant

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Abstract

We present an iterative refinement of the UQFF U_b (buoyancy) model across 35 sequential Kepler Orrery V animation frames (22 September – 27 October 2011). Each frame updates the environmental force F_env(t) = w_orbit*F_orbit + w_tide*F_tide + w_gal*F_gal with orbital phase perturbations. The force sub-equations are: F_orbit = G*M_p*M_s/a^3 (orbital gradient), F_tide = G*M_p*M_s*R_p/a^6 (tidal), F_gal = v_gal^2/r_gal + G*M_DM/r_gal^2 (galactic dark matter). Convergence to F_env ~ 6.5e-2 m/s^2 validates the U_b model for the Earth-Sun system and extends to 1,200+ Kepler exoplanet candidates.

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1. Core Equations

- `F_env(t) = w_orbit*F_orbit + w_tide*F_tide + w_gal*F_gal`
- `F_orbit = G*M_p*M_s / a^3`
- `F_tide = G*M_p*M_s*R_p / a^6`
- `F_gal = v_gal^2/r_gal + G*M_DM/r_gal^2`
- `Weights: w_orbit=0.5, w_tide=0.3, w_gal=0.2`
- `Convergence: F_env ~ 6.5e-2 m/s^2 across 35 frames`

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2. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

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3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
- **Session:** 199
- **VDS/DVP/BH:** ABSENT (general vacuum density references only)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging

5. UQFF Calibration: kappa=0.0005/day, [SSq]=0.57, beta_i~0.603